

Harshana Lakshara Fernando

Software Engineer | DevOps & Cloud Engineering

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Summary

Software Engineer with hands-on experience in Kubernetes, Docker, AWS, CI/CD, Kafka, Linux-based deployments, monitoring, and production support. Experienced in deploying distributed systems, automating delivery pipelines, troubleshooting production environments, and improving system reliability. CKA-trained, with strong Java, Spring Boot, and cloud-native engineering experience.

Skills

Cloud & DevOps	AWS Kubernetes Docker Docker Compose CI/CD Jenkins GitHub Actions NGINX Traefik
Infrastructure & Operations	Linux Bash/Shell Scripting Networking Reverse Proxy SSL/TLS Production Support
Monitoring & Testing	Prometheus Grafana JMeter Gatling Logging Performance Testing
Messaging & Distributed Systems	Kafka RabbitMQ gRPC WebSockets Protobuf Resilience4j
Backend	Java Spring Boot Spring Cloud Go Gin Node.js REST APIs
Databases	PostgreSQL Oracle MySQL Redis Couchbase DynamoDB

Experience

R&D Software Engineer — KōreroCare Platform

Mar 2026 – Present

Empathic Computing Laboratory, University of Auckland / Remote

- Containerized backend, database, authentication, and gateway services using **Docker Compose**.
- Configured **Traefik** routing for internal services and integrated **Keycloak OAuth2/OIDC** authentication.
- Managed environment configuration, service networking, health checks, and local deployment workflows.
- Designed **PostgreSQL**-backed backend services for high-volume mobile and wearable sensor ingestion.
- Supported secure service-to-service communication and deployment documentation for the research team.

Software Engineer Intern — Information Systems Associates (Pvt) Ltd

Dec 2024 – May 2025

Aviation Technology / Colombo, Sri Lanka

- Deployed and supported a production Spring Boot and Kafka distributed service handling **25+ requests per second** with **<100ms latency**.
- Built Jenkins-based environment support and coordinated releases across development, QA, staging, and production environments.
- Created JMeter load tests and Prometheus/Grafana monitoring to track service latency, Kafka consumer behaviour, queue times, and order-processing performance.
- Diagnosed and resolved **Kafka consumer lag**, offset-reset, retry, and graceful-shutdown issues in production.
- Integrated **gRPC** and **Kafka** communication between a legacy Java 8 monolith and newer microservices.
- Supported phased production migration using feature toggles, rollback mechanisms, and a **strangler-pattern** deployment strategy.
- Maintained automated tests with **>90% unit test coverage** and collaborated with QA and distributed engineering teams.

Freelance Software Engineer — Selected Client Projects

May 2025 – May 2026

Client Platforms / Remote / Sri Lanka

- Delivered Agrinova, a full-stack multi-branch inventory platform deployed on **Vercel** and **Supabase/PostgreSQL**, managing transaction-safe stock logic across 3 core workflows (issuance, returns, purchases).
- Reduced repeat dashboard data-fetch latency from ~384ms to <1ms for cached views using **React Query** caching and pagination. Live demo: agrinova-demo.vercel.app.

- Owned end-to-end client-facing fixes, resolving production authentication, version compatibility, and workflow-level validation bugs.

Projects

See more projects on my portfolio

1. Employee Management System (EMS) (2025 | Individual) Architecture | Github

Designed and built a comprehensive, cloud-native microservices application for employee and department management.

- Containerized Spring Boot microservices and deployed them using **Kubernetes manifests** for deployments, services, configuration, and service discovery.
- Implemented CI/CD workflows using **GitHub Actions** and **ArgoCD**, with centralized configuration, API Gateway routing, Prometheus/Grafana monitoring, and Resilience4j fault tolerance.

Keywords: Spring Boot, Spring Cloud, Kubernetes, Docker, GitHub Actions, ArgoCD, Prometheus, Grafana, Resilience4j, API Gateway

2. Serverless Auction Platform (2025 | Individual) Architecture | Github

A capstone project built using AWS, following a microservices architecture with an event-driven, serverless (FaaS) design.

- Designed an AWS serverless architecture using **Lambda**, **API Gateway**, **DynamoDB**, **EventBridge**, **SQS**, and **SNS**, with asynchronous event processing and centralized error handling.
- Utilized **LocalStack** and **Docker** for local cloud emulation, ensuring reliable testing and deployment workflows.

Keywords: Node.js, Serverless Framework, AWS Lambda, DynamoDB, SQS, SNS, EventBridge, LocalStack, Docker

3. GoTogether (2024 | Group) Architecture | Live | Github

Developed and deployed a full-stack travel assistant platform with microservices in Next.js, Spring Boot, and Go.

- Deployed containerized Next.js, Spring Boot, Go, Keycloak, and PostgreSQL services behind **NGINX** on **Oracle Cloud**, with **Certbot**-managed HTTPS.
- Built a dedicated Auth Service using **Keycloak**, managing OAuth2/OIDC flows, token issuance, and automated realm/client boot-time import scripts.

Keywords: Next.js, Go, Spring Boot, Keycloak, PostgreSQL, NGINX, Docker, Oracle Cloud, Certbot

4. Event-Driven Food Delivery Platform (2024 | Group) Github

Built a scalable, event-driven food delivery platform using a modern microservices architecture and streaming.

- Enabled robust order synchronization using **Kafka**, processing asynchronous ‘order_created’ events across distributed microservices.
- Integrated **Redis caching** to improve API performance and reduce PostgreSQL database load, deploying the stack securely via **Docker**.

Keywords: Java, Spring Boot, Kafka, Redis, PostgreSQL, Docker, Microservices, Event-Driven

5. ZK-Rollup L2 Benchmarking Framework

Research Project — University of Moratuwa

- Built Docker-based benchmark orchestration scripts to evaluate Ethereum Layer-2 trade-offs across sequencing, batching, and data availability.
- Implemented the Rust-based Submitter service, observing **70% lower DA publication cost** using EIP-4844-style blobs versus calldata.
- Introduced mock-testable clients to improve testability and increased verifiable test coverage from **41% to 88%**.

Keywords: Rust, Docker, Python, L2 Rollups, Benchmarking

Education

B.Sc. (Hons) in Engineering – Computer Science and Engineering

2022 – 2026

University of Moratuwa, Sri Lanka

G.C.E. Advanced Level (Physical Science Stream)

2019

St. John's College, Panadura, Sri Lanka

Results: 3 A's (Combined Mathematics, Physics, Chemistry)

District Rank: 17

Certifications & Professional Training

Certified Kubernetes Administrator preparation — KodeKloud, 2025 • AWS Solutions Architect Associate preparation — Udemy, 2025 • 100 Days of DevOps — KodeKloud, 2025 • Apache Kafka Series — Udemy, 2025 • Advanced OAuth Security — Udemy, 2024